Md. Jubaer Khan

Senior Software Engineer — Embedded Systems & C/C++

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PROFESSIONAL SUMMARY

Senior Software Engineer with over five years of experience delivering embedded, desktop and web software, including four-plus years at BJIT Ltd. building safety-critical systems for Japanese and international clients. Strong in C and C++ across RTOS and Embedded Linux targets, automotive and industrial communication protocols (CAN 2.0B, SAE J1939), and MISRA-C / MISRA-C++ / BARR-C compliant development. Experienced in requirement analysis, system architecture, code review and mentoring within Agile/Scrum teams. Currently leads firmware for a cashless credit gateway on Arm Cortex-M; recent work also applies AI agents across the software development life cycle and modernises legacy VB6 systems to web-based C#/.NET. Recognised as BJIT Best Performer of the Year for 2023–2024, and co-author of a peer-reviewed IEEE Transactions paper on deep-learning-based COVID-19 detection.

TECHNICAL SKILLS

Programming Languages	C, C++ (Embedded C/C++, Visual C++), Python, C#, VB.NET, Java (core), JavaScript
Embedded & RTOS	RTOS (FreeRTOS), lwIP TCP/IP stack, Embedded Linux (Yocto, Buildroot), STM32 and ESP32/ESP8266 ecosystems, embedded system design, peripheral and sensor interfacing, schematic and datasheet analysis
Protocols	CAN 2.0B, SAE J1939, RS-232, UART, SPI, I2C, TCP/IP, Ethernet, RFID, Wi-Fi (IEEE 802.11)
Standards	MISRA-C, MISRA-C++, BARR-C, design patterns, UML, OOAD, test-driven development, code and design review
Desktop & Web	Qt Framework, Flutter/Dart, MFC, .NET Framework / .NET Core, ASP.NET MVC, Razor, Flask, Django, HTML, CSS, Bootstrap, MySQL, MS SQL Server
AI & ML	AI agents, RAG, MCP servers, LLM orchestration, human-in-the-loop, ML/NLP (TensorFlow, CNN)
Tools & Process	Git, GitHub, Bitbucket, Jira, Confluence, Visual Studio, VS Code, PlatformIO, CodeWarrior, Orchestra Suite, OpenTelemetry, Grafana, Prometheus, Loki, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer *Jul 2025 – Present*

BJIT Ltd. • Dhaka, Bangladesh

- Provide technical leadership for embedded and PC application projects, owning system architecture and critical modules.
- Lead firmware development on a cashless credit gateway for cash-handling terminals, from protocol analysis to a multi-vendor proof of concept.
- Lead design and code reviews, enforcing MISRA-C, MISRA-C++ and BARR-C compliance across project teams.
- Mentor junior engineers through design walkthroughs, technical guidance and best-practice sharing.
- Resolve complex system-level defects in RTOS and Embedded Linux environments and support integration phases.

Software Engineer *Apr 2022 – Jun 2025*

BJIT Ltd. • Dhaka, Bangladesh

- Designed, implemented, tested and deployed embedded, desktop and web software components for client products.
- Developed communication-driven applications over CAN 2.0B, SAE J1939, UART, SPI and I2C.
- Built custom Embedded Linux distributions with Yocto and Buildroot for target hardware platforms.
- Delivered cross-platform desktop applications in C++/Qt and PC applications on .NET (C#, VB.NET).

Web Developer & Designer (part-time) *Nov 2021 – Feb 2022*

Independent client engagement

- Planned, built, deployed and maintained a client business website with WordPress and cPanel, including professional email setup.

Intern Software Engineer *Feb 2021 – Jun 2021*

North South University — Office of Information Technology

- Built an NLP-based student assistance chatbot in Python and Flask, deployed live on university infrastructure.
- Developed an RFID-based automated printing and access facility using C++, ESP32 and the RC522 reader over SPI.

Student Worker, IT Support Team *Aug 2018 – Jan 2021*

North South University — Office of Information Technology

- Provided technical support to faculty, officials and students, and staffed the IT service desk and request tracking.
- Prepared and maintained PCs for printing and engineering labs, covering OS and software installation, IP and domain setup.

SELECTED PROFESSIONAL PROJECTS

Client and product names are withheld under confidentiality agreements.

Cashless Credit Gateway for Cash-Handling Terminals
Embedded firmware for a bridge board integrating cash-handling terminal hardware with a merchant point-of-sale system, plus a desktop simulator used to validate PC integration. **Tech:** *C, Arm Cortex-M (STM32), FreeRTOS, lwIP, TCP/IP, Ethernet, RS-232/UART, Flutter/Dart* • *Team lead*

In-House R&D — Agentic SDLC & AI Agent-Based Monitoring
Two in-house R&D initiatives on AI agents: one applied across the software development life cycle, one for infrastructure monitoring and incident response. **Tech:** *AI agents, RAG, MCP, Python, OpenTelemetry* • *In-house R&D*

Legacy Management Software Modernisation — VB6 to C#/.NET
Migration of a legacy desktop management application to a modern web stack, using AI-assisted migration tooling. **Tech:** *VB6, C#, .NET 8, Blazor Server, Radzen, ActiveReports, PostgreSQL, GitHub Copilot* • *Team of 15*

Vehicle Controller Migration — Turret & Boom Control
Safety-critical controller application for automotive-grade vehicle systems, with an off-vehicle CAN simulator for testing. Owned one part of the overall project and gave the technical knowledge transfer needed by newly onboarded engineers. **Tech:** *C, C++, CAN 2.0B, SAE J1939, NXP Power Architecture MCU, CodeWarrior, PCAN* • *Team of 5*

Robotics R&D — Robotic Arm & Spot Sensor PoCs
Two robotics proofs of concept: an industrial robotic-arm digital twin with kinematics and multi-arm control, and custom-sensor feasibility on a quadruped platform. **Tech:** *Unity (C#), C/C++, Arduino, Python, gRPC, CAN* • *In-house R&D*

Custom Embedded Linux Distribution
Custom Embedded Linux board support and image builds for an Arm single-board computer, covering toolchain, bootloader, device tree and BSP layer. **Tech:** *Yocto, Buildroot, U-Boot, Device Tree, TI AM335x, Linux* • *Team of 2*

Cross-Platform Desktop Application (Qt)
Cross-platform log-viewer application and reusable logging library with multithreaded loading, filtering and search. **Tech:** *C++, Qt 6, CMake, GoogleTest* • *Team of 2*

Legacy Vehicle Interface Module Firmware Migration
Firmware migration of a legacy vehicle interface module to a modern rugged controller platform, with off-target test builds and hardware validation. **Tech:** *C, C++, CAN 2.0B, SAE J1939, Nucleus RTOS, NXP Power Architecture MCU, CodeWarrior, SCons, Conan, Python, PCAN* • *Team of 6*

EDUCATION

B.Sc. in Computer Science & Engineering *May 2017 – May 2021*
North South University, Dhaka, Bangladesh
Graduated Cum Laude • *Specialisation coursework: Machine Learning, Pattern Recognition & Neural Networks, Theory of Computation, Mobile & Wireless Application Development*

Higher Secondary Certificate (HSC), Science *2013 – 2015*
Jhalokathi Govt. College, Jhalokathi, Bangladesh

RESEARCH & PUBLICATIONS

XCR-Net: A Computer Aided Framework to Detect COVID-19 *2024*
Ashik Mostafa Alvi, **Md. Jubaer Khan**, Nishat Tasnim Manami, Zubair Azim Miazi, Kate Wang, Siuly Siuly, Hua Wang • *IEEE Transactions on Consumer Electronics, vol. 70, no. 4, pp. 7551–7561* • doi.org/10.1109/TCE.2024.3446793

Building COVID Knowledge Base from Medical Journals *2021*
Md. Jubaer Khan, Walidul Alam Ehab • *Undergraduate thesis, Dept. of Electrical & Computer Engineering, North South University* • repository.northsouth.edu

AWARDS & ACHIEVEMENTS

Best Performer of the Year — BJIT Ltd. *2023–2024*

LANGUAGES & ADDITIONAL INFORMATION

Training	Embedded Systems engineering course, BJIT Academy (2022) • AI Fundamentals, IBM SkillsBuild (2026) • Information Security IS-101, BJIT Academy (2024)
Languages	Bangla — native • English — fluent (reading, writing, speaking)
Personal	Date of birth 13 December 1998 • Nationality: Bangladeshi

References available on request.